

Sonifying the Heart: Inner Emotion Detection from ECG through Music

Music Workshop Sprint 3: Individual Contribution

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1 My Role

I built the first two stages of our pipeline: **ECG data** and **Heartbeats**. Every clip that the music and emotion stages use comes from my part.

2 What I Did

- **Data access.**
 - Read `afdb` and `mitdb` straight from PhysioNet, streaming only the annotations and 30 s of signal.
 - Wrote a parallel downloader for the full databases, because PhysioNet limits each connection to about 20 KB/s.
- **Annotations.** Handled both formats: `afdb`'s separate beat (`.qrs`) and rhythm (`.atr`) files, and `mitdb`'s single file that mixes beat labels and rhythm marks.
- **Heartbeat measures.** R–R intervals, heart rate and RMSSD for every clip.
- **Clip selection.** For each recording, one 30 s normal clip (the steadiest of up to five windows) and one 30 s AFib clip (the window with the most detected beats).
- **Records.**
 - Chose five `afdb` recordings and `mitdb` 201 and 202.
 - Excluded `mitdb` 219. Its “normal” stretch (RMSSD 270.6 ms) is more irregular than its AFib (145.9 ms), because of 133 blocked beats.
- **Any ECG file.**
 - Added XQRS heartbeat detection.
 - On record 04043 it found 53 beats, a mean R–R of 559 ms and an RMSSD of 14.2 ms. The expert annotations give 54 beats, 559 ms and 13.8 ms.

3 Key Result

`afdb`'s beat marks come from an automatic detector and jitter in places. Taking the first 30 s made two normal clips look more irregular than AFib (record 04015: 199 ms against 105 ms).

After switching to the steadiest window, every recording separates clearly:

- normal clips: RMSSD 7.6–64.1 ms;
- AFib clips: RMSSD 99.9–194.3 ms, 3–17 times the same recording's normal value.

4 Next Steps

- Load the ECG of emotion-labelled datasets (WESAD, DREAMER, AMIGOS) to validate the emotion stage.
- Add the Sprint 2 items still open: SDNN, pNN50, Poincaré plots and rate-matched clips.